Report
Of

Internship Report on Java Full Stack Development:

"Employee Management System"

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: 2410030202
Name of Student: AMIT YADAV
Batch:
2026-27

CANDIDATE'S DECLARATION

I, **Amit Yadav**, Roll No. **2410030202**, do hereby declare that the internship report titled ***"Employee Management System"*** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Amit Yadav

Roll No.: 2410030202

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program on Java Full Stack Development conducted by EduSkills Academy. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement throughout the duration of this internship and the preparation of this report.

I express my gratitude to my parents for their blessings.

Signature

Student Name: Amit Yadav

Roll No.: 2410030202

INTERNSHIP COMPLETION CERTIFICATE

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EduSkills
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AICTE - EduSkills
2026
VIRTUAL INTERNSHIP

Certificate of Virtual Internship

This is to certify that

Amit Yadav

IILM University, Greater Noida

has successfully completed the 8-weeks

Java Full Stack Developer Virtual Internship

During June - August 2026

Supported by EduSkills ACADEMY

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills

Certificate ID : 47c99050c8078c949c38

Student ID: STU6997dc3d1dcdc1771559997

INTERNSHIP GRADE
O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents the work carried out during an 8-week Virtual Internship on Java Full Stack Development, undertaken with EduSkills Academy under the AICTE Internship Scheme. The internship provided a structured, hands-on learning path covering the complete web development lifecycle, beginning with front-end fundamentals such as HTML and web page structure, progressing through CSS and Bootstrap for responsive design, JavaScript and jQuery for interactive client-side behaviour, and culminating in server-side development using Core and Advanced Java, the Spring Framework, Spring Boot, Hibernate ORM, MySQL and Git version control.

As part of the internship, a capstone project titled "Employee Management System" was designed and developed to consolidate and demonstrate the skills acquired during the program. The project is a web-based application that allows an organization to digitally manage employee records, departments, attendance and basic payroll information, replacing manual, paper-based record keeping with a centralised, searchable and secure system.

4.2 Organization Profile

EduSkills Academy is the training arm of the EduSkills Foundation, a not-for-profit, social ed-tech organisation headquartered in Bhubaneswar, Odisha. The organisation works in partnership with the All India Council for Technical Education (AICTE) and various industry and technology partners to deliver Virtual Internship programs aimed at building an Industry 4.0-ready digital workforce. Its stated mission centres on "learning for employability" — bridging the gap between academic curricula and the practical, job-ready skills expected by the technology industry.

Through the AICTE Internship Portal, EduSkills Academy offers structured, self-paced virtual internship tracks of typically eight weeks' duration across domains such as Java Full Stack Development, Artificial Intelligence and Machine Learning, Cloud Computing, Cybersecurity and Data Analytics. Each track combines guided learning modules, hands-on assignments and a final capstone project, and concludes with an industry-recognised completion certificate, which is co-branded with the relevant technology partner.

4.3 Problem Statement

Many small and medium-sized organisations continue to rely on manual registers or disconnected spreadsheets to maintain employee information such as personal details, department allocation, attendance and salary records. This manual approach is time-consuming, error-prone, difficult to search, and offers no centralised way to generate reports or enforce data consistency. There is a clear need for a simple, web-based Employee Management System that allows HR staff to add, update, search and manage employee data through a secure and user-friendly interface, backed by a relational database.

4.4 Project Objectives

- To design and develop a web-based system for managing employee records, departments and attendance.
- To implement a responsive and user-friendly front-end using HTML, CSS, Bootstrap, JavaScript and jQuery.
- To build a robust back-end using Core Java, the Spring Framework and Spring Boot that exposes RESTful services for CRUD (Create, Read, Update, Delete) operations.
- To use Hibernate ORM for mapping Java objects to relational database tables and to use MySQL as the underlying database.
- To apply Git for source code version control throughout the development process.
- To validate the functionality of the system through systematic testing of its core modules.

4.5 Scope of the Project

The Employee Management System covers the core functions required to maintain employee data within a small organisation. Its scope includes:

- Employee registration, profile update and record deletion (CRUD operations).

- Department creation and assignment of employees to departments.
- Daily attendance marking and viewing of attendance history for each employee.
- Basic payroll information storage such as designation and monthly salary.
- Search and filter functionality to locate employee records quickly.
- Role-based access so that only authorised users (HR/Admin) can modify records.

The system is designed as a foundational academic prototype; it does not cover advanced enterprise features such as biometric attendance integration, statutory payroll compliance (PF/ESI/Tax) or multi-branch organisational hierarchies, which are identified as areas for future enhancement.

4.6 Technologies and Tools Used

The project follows a layered, full-stack Java architecture. Table 4.1 summarises the technologies used at each layer of the application.

Layer	Technology Used	Purpose
Front-End	HTML5, CSS3, Bootstrap	Structuring and styling the responsive user interface
Front-End Scripting	JavaScript, jQuery	Client-side interactivity, form validation and DOM manipulation
Back-End Language	Core Java, Advanced Java	Implementation of business logic and server-side processing
Back-End Framework	Spring Framework, Spring Boot	Dependency injection, REST API development and rapid application setup
Persistence / ORM	Hibernate	Object-Relational Mapping between Java objects and database tables
Database	MySQL	Storage and retrieval of employee, department and payroll records
Version Control	Git	Source code versioning and collaborative development

Table 4.1: Technology Stack Summary

In addition to the above, the following development tools were used: Eclipse / IntelliJ IDEA as the Integrated Development Environment, Apache Maven for project build and dependency management, Postman for testing REST APIs, MySQL Workbench for database design and management, and Git together with GitHub for source code version control and collaboration.

4.7 System Architecture

The Employee Management System follows a layered (n-tier) architecture that separates presentation, business logic and data persistence concerns, in line with standard Spring-based full-stack application design. Figure 4.1 illustrates the overall architecture.

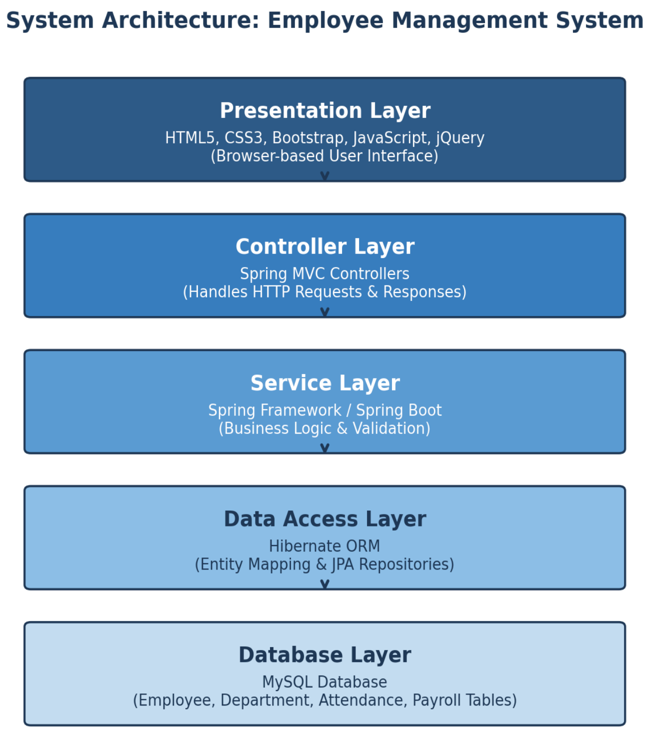

Figure 4.1: System Architecture of Employee Management System

The Presentation Layer, built with HTML, CSS, Bootstrap, JavaScript and jQuery, renders the user interface in the browser and communicates with the server through HTTP requests. The Controller Layer, implemented with Spring MVC, receives these requests, delegates processing to the Service Layer, and returns appropriate responses. The Service Layer, built on the Spring Framework and Spring Boot, contains the core business logic and validation rules. The Data Access Layer uses Hibernate ORM to map Java entity classes to relational tables and to perform database operations without writing raw SQL for common cases. Finally, the Database Layer, powered by MySQL, persistently stores all employee, department, attendance and payroll data.

4.8 Methodology

The project was developed using an iterative and incremental approach, broadly aligned with Agile practices, over the course of the eight-week internship. The methodology followed these key stages:

- Requirement Analysis: Identifying the core modules — Employee, Department, Attendance and Payroll — and defining the functional requirements for each.
- Design: Preparing the database schema (ER diagram), designing REST API endpoints, and planning the layered system architecture shown in Figure 4.1.
- Front-End Development: Building responsive HTML/CSS/Bootstrap pages for employee registration, listing, search and attendance, enhanced with JavaScript/jQuery for client-side validation and AJAX-based interactions.
- Back-End Development: Implementing Spring Boot REST controllers, service classes containing business logic, and Hibernate-based repository/DAO classes mapped to MySQL tables.
- Integration: Connecting the front-end pages to the back-end REST APIs and verifying end-to-end data flow from the browser to the database and back.
- Testing: Performing unit and functional testing of individual modules (see Table 4.2) using Postman for API testing and manual testing of the user interface.
- Version Control: Committing code incrementally to a Git repository, with meaningful commit messages tracking each stage of development.

Table 4.2 summarises representative test cases executed to validate core functionality.

Test ID	Test Case	Expected Result	Status
TC-01	Add a new employee record with valid data	Record saved and displayed in employee list	Pass
TC-02	Add an employee record with a duplicate employee ID	System rejects the entry with a validation error	Pass
TC-03	Update an existing employee's department	Updated department reflected immediately in the list	Pass
TC-04	Delete an employee record	Record removed from the database and UI	Pass
TC-05	Mark attendance for an employee for the current date	Attendance entry stored against correct employee and date	Pass
TC-06	Search for an employee by name	Matching record(s) displayed correctly	Pass

Table 4.2: Test Cases and Results

4.9 Expected Outcomes

On successful completion of the project, the Employee Management System is expected to deliver the following outcomes:

- A fully functional web application that allows HR staff to manage employee, department and attendance data through a single, centralised interface.
- Reduction in manual, paper-based record-keeping errors and faster retrieval of employee information through search and filter features.
- A working demonstration of REST API design and consumption using Spring Boot, showcasing separation of concerns between front-end and back-end.
- Practical, hands-on experience in integrating the technologies covered during the internship — HTML, CSS, Bootstrap, JavaScript, jQuery, Core/Advanced Java, Spring, Spring Boot, Hibernate, MySQL and Git — into a single cohesive, full-stack application.
- A reusable project foundation that can be extended in future with features such as role-based authentication, payroll automation and reporting dashboards.

4.10 Certificate of Completion

The internship was successfully completed, and a Certificate of Internship was issued by EduSkills Academy on August 08, 2026 (Certificate ID: 2026-671922F184), confirming completion of the 8-week Java Full Stack program covering HTML, CSS, Bootstrap, JavaScript, jQuery, Core Java, Advanced Java, Spring Framework, Spring Boot, Hibernate, MySQL and Git & Version Control. A copy of the certificate is attached in this report under "Internship Completion Certificate" (page iii).

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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